

Intro to Table Relations in SQL

 [_ \(https://github.com/learn-co-curriculum/python-p3-intro-to-sql-table-relations\)](https://github.com/learn-co-curriculum/python-p3-intro-to-sql-table-relations)  [_ \(https://github.com/learn-co-curriculum/python-p3-intro-to-sql-table-relations/issues/new\)](https://github.com/learn-co-curriculum/python-p3-intro-to-sql-table-relations/issues/new)

Learning Goals

- Use primary and foreign keys to create the relations in a relational database.
- Use **INNER JOIN** and **LEFT JOIN** to retrieve relevant data for members of one table in other tables.
- Use joins to create one-to-many and many-to-many relationships between tables.

Key Vocab

- **Primary Key**: a number that uniquely identifies one record in a table.
- **Foreign Key**: a column or group of columns that connects one table to another.
- **Join**: a query that returns related records from multiple tables in a single record.
- **One-to-Many**: a type of relationship between tables where one record in table A is connected to multiple records in table B. **e.g.** One person ordering multiple drinks at a bar.
- **Many-to-Many**: a type of relationship between tables where multiple records in table A are connected to multiple records in table B. **e.g.** Students have many classes and classes have many students.

Introduction

The majority of databases you'll work with as a developer will have more than one table, and those tables will be connected together in various ways to form table relationships.

In this section we'll cover:

- The structure of a relational database as tables
- Primary key and foreign keys
- How to associate data tables using a foreign key column
- How to model **one-to-many** and **many-to-many** relationships
- How a join table is used
- Different types of joins

Over the next several lessons, we'll discuss having multiple tables in a database, how to define relationships between different tables, and explain the different types of table relationships that can exist.

Resources

- [What is a Relational Database? - Google Cloud](https://cloud.google.com/learn/what-is-a-relational-database) ➞ (https://cloud.google.com/learn/what-is-a-relational-database)
- [Difference between Primary Key and Foreign Key - GeeksforGeeks](https://www.geeksforgeeks.org/difference-between-primary-key-and-foreign-key/) ➞ (https://www.geeksforgeeks.org/difference-between-primary-key-and-foreign-key/)
- [SQL Joins - W3Schools](https://www.w3schools.com/sql/sql_join.asp) ➞ (https://www.w3schools.com/sql/sql_join.asp)
- [Airtable's guide to many-to-many relationships](https://support.airtable.com/hc/en-us/articles/218734758-Airtable-s-guide-to-many-to-many-relationships) ➞ (https://support.airtable.com/hc/en-us/articles/218734758-Airtable-s-guide-to-many-to-many-relationships)
- [Practice SQL Queries on SQLBolt](http://sqlbolt.com/lesson/select_queries_review) ➞ (http://sqlbolt.com/lesson/select_queries_review)
- [dbdiagram.io: ERD Drawing Tool](https://dbdiagram.io) ➞ (https://dbdiagram.io)